

SOUTHERN CALIFORNIA, CA, USA

WMO: 723825

Lat: 34.583N

Lon: 117.383W

Elev: 2885

StdP: 13.23

Time Zone: -8.00 (NAP)

Period: 86-17

WBAN: 23131

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.5	28.3	-0.5	5.9	44.2	1.9	6.7	43.5	28.0	47.1	25.1	49.6	2.5	170	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.6	102.0	64.2	99.8	63.1	97.6	62.4	68.2	89.2	66.8	89.9	65.6	90.5	9.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	94.9	75.1	59.1	83.1	75.8	55.5	73.1	76.8	34.0	87.3	32.9	89.6	31.8	90.5	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	21.3	18.7	DB	20.1	107.1	4.7	2.3	16.7	108.8	14.0	110.1	11.3	111.4	7.9	113.1	(4)
(5)			WB	16.4	71.5	4.1	3.0	13.5	73.7	11.1	75.4	8.8	77.1	5.8	79.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	45.7	50.4	55.2	61.7	67.4	77.2	82.2	81.5	75.1	65.7	53.5	44.8		
	DBStd	14.57	5.76	7.57	6.74	6.68	7.39	6.68	5.17	4.17	6.85	7.49	7.04	6.52		
	HDD50	496	155	81	27	4	0	0	0	0	0	2	44	183		
	HDD65	2596	600	409	308	139	64	6	0	0	15	82	348	625		
	CDD50	5400	20	93	189	356	538	816	998	976	752	490	149	23		
	CDD65	2025	0	1	5	40	137	372	533	511	317	105	4	0		
	CDH74	27090	0	36	183	711	1863	4990	7193	6697	3884	1432	100	1		
	CDH80	14995	0	4	27	219	795	2902	4393	4015	2053	575	12	0		
(14) Wind	WSAvg	6.4	5.0	6.4	7.7	8.0	8.6	7.8	6.7	6.1	5.6	5.1	4.9	5.0		
(15) Precipitation	PrecAvg	5.3	0.9	1.5	0.8	0.2	0.2	0.1	0.2	0.2	0.2	0.4	0.3	0.7		
	PrecMax	10.9	5.5	5.6	3.0	1.1	1.6	0.8	1.0	1.1	1.5	3.8	1.1	3.3		
	PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.1	1.5	1.6	1.0	0.4	0.4	0.2	0.2	0.3	0.4	0.9	0.4	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.7	79.1	83.5	90.0	95.2	106.1	106.9	102.5	99.1	93.8	81.7	70.4		
		MCWB	49.3	52.3	53.9	57.5	58.9	63.4	65.9	65.2	63.1	60.2	54.8	50.1		
	2%	DB	64.8	73.3	79.2	85.9	91.4	100.3	102.4	100.3	96.7	90.4	76.9	64.3		
		MCWB	46.7	49.8	52.4	56.3	58.4	61.9	64.5	64.0	62.0	58.6	52.1	47.1		
	5%	DB	62.6	70.0	74.9	82.0	88.6	98.4	100.1	99.1	93.3	87.0	72.6	61.4		
		MCWB	45.7	48.6	51.2	54.0	57.9	61.5	63.6	63.2	61.2	57.0	50.3	45.2		
	10%	DB	58.4	65.6	71.5	78.2	84.2	94.0	98.1	96.8	91.1	82.1	68.1	57.4		
		MCWB	44.5	47.2	50.4	52.6	56.9	60.4	62.9	62.9	60.7	55.0	49.6	43.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	55.3	56.4	60.0	64.7	66.3	69.7	72.1	67.6	63.4	57.7	53.4		
		MCDB	60.5	71.8	75.2	83.3	84.8	92.6	91.2	86.8	86.5	75.6	72.3	61.1		
	2%	WB	51.3	52.6	54.4	57.2	61.7	64.5	68.0	68.9	66.1	61.0	55.1	51.2		
		MCDB	60.4	64.6	73.4	81.4	82.1	93.9	90.2	88.1	86.5	79.7	69.0	59.4		
	5%	WB	48.7	50.6	52.8	55.4	59.6	63.3	66.7	67.1	64.5	59.2	52.9	48.7		
		MCDB	56.6	63.9	71.1	78.0	82.7	93.5	90.7	90.2	86.6	79.9	68.3	56.8		
	10%	WB	46.2	48.8	51.1	53.8	57.6	61.7	65.5	65.6	62.9	57.5	50.8	45.6		
		MCDB	55.3	62.4	67.9	74.5	82.0	90.1	91.9	90.7	86.0	78.8	65.1	55.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.0	24.2	25.4	27.2	27.9	30.6	29.6	29.7	28.5	27.9	25.8	22.3		
		MCDBR	28.5	30.0	31.6	32.3	31.5	33.5	31.7	31.4	30.4	31.0	30.6	27.1		
		MCWBR	15.6	14.1	13.7	13.2	11.5	11.5	11.2	10.5	10.6	12.8	15.1	14.9		
	5% WB	MCDBR	21.9	24.7	28.2	29.2	28.8	30.6	26.5	25.8	26.5	26.0	25.7	21.8		
		MCWBR	13.1	12.5	12.9	12.5	11.2	11.2	10.2	10.3	9.8	11.1	13.1	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.275	0.291	0.309	0.318	0.320	0.345	0.328	0.312	0.298	0.269	0.269		
	taud		2.577	2.569	2.507	2.456	2.458	2.471	2.453	2.489	2.517	2.534	2.601	2.589		
	Ebn at Noon		305	312	313	310	306	304	295	300	302	299	301	297		
	Edh at Noon		24	27	32	35	36	35	36	34	31	28	23	23		
(44) All-Sky Solar Radiation	RadAvg		1025	1325	1795	2204	2469	2667	2473	2325	2002	1535	1151	932		
	RadStd		74	123	83	80	95	105	115	75	54	76	45	60		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	+1.19	N/A	-3.26	+1.24	N/A	N/A	N/A	-212	+421	+214

Nomenclature: See separate page